

Customer Segmentation Model Customer Personality Analysis & Segmentation

Module 6 | ML-M06 | Final Project Report

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Assigned To	Machine Learning Intern Team
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Difficulty Level	Intermediate
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Dataset	marketing_campaign.csv (2,240 records, 29 columns)

1. Executive Summary

This project applies unsupervised machine learning to the Customer Personality Analysis dataset in order to identify meaningful customer segments that can support marketing strategy, retention, and personalized recommendations. The pipeline covers the full lifecycle: data validation, feature engineering, feature selection, scaling, exploratory analysis, optimal cluster determination, training and evaluation of four clustering algorithms (K-Means, Hierarchical, Gaussian Mixture, DBSCAN), cluster profiling, stability testing, and packaging of a production-ready pipeline.

Best performing algorithm: K-Means (K = 2) — Silhouette Score = 0.2985, Davies-Bouldin Index = 1.3738.

2. Dataset Overview

Metric	Value
Rows	2,240
Original Columns	29
Columns after cleanup (ID removed)	28
Missing Values (Income)	24 (filled with median)
Duplicate Records	0

3. Feature Engineering

New features were derived from the raw dataset to better represent customer behaviour:

- Customer_Age = 2026 - Year_Birth
- Total_Spending = sum of spending across Wines, Fruits, Meat, Fish, Sweets, Gold products
- Total_Children = Kidhome + Teenhome
- Family_Size = Total_Children + 2
- Purchase_Frequency = Web + Catalog + Store purchases
- Accepted_Campaigns = sum of AcceptedCmp1-5 + Response
- Customer_Tenure = days between Dt_Customer and reference date (2026-01-01)

4. Feature Selection & Scaling

12 features were selected for clustering: Customer_Age, Income, Total_Spending, Recency, Customer_Tenure, Family_Size, Total_Children, Purchase_Frequency, Accepted_Campaigns, NumWebPurchases, NumStorePurchases, NumCatalogPurchases.

Correlation check: Total_Children and Family_Size were found to be perfectly correlated ($r = 1.00$), as expected since Family_Size is derived directly from Total_Children.

Both StandardScaler and MinMaxScaler were fitted and saved; the standardized dataset was used for all subsequent clustering.

5. Exploratory Cluster Analysis

Distribution plots, boxplots, a pair plot, and correlation heatmaps were generated for all selected features. A 2-component PCA explained 41.1% (PC1) and 13.8% (PC2) of total variance. IQR-based outlier detection flagged the highest counts in Accepted_Campaigns (97), Family_Size / Total_Children (53 each), and NumCatalogPurchases (23).

6. Determining the Optimal Number of Clusters

6.1 Elbow Method (WCSS)

K	WCSS
2	18219.04
3	16239.28
4	14584.11
5	13615.54
6	12978.56
7	12466.37
8	12063.29
9	11619.04
10	11073.17

Suggested Optimal K (elbow-based heuristic): **3**

6.2 Silhouette Score vs Davies-Bouldin Index

K	Silhouette	Davies-Bouldin
2	0.2985	1.3738
3	0.2245	1.7876
4	0.1895	1.7779
5	0.1916	1.7886
6	0.1766	1.8365
7	0.1492	1.9404
8	0.1471	1.9780
9	0.1541	1.7362
10	0.1534	1.7267

Best K by Silhouette Score: **2** (0.2985). Best K by Davies-Bouldin Index: **2** (1.3738). Both metrics agree that K = 2 provides the most compact and well-separated clustering.

7. Clustering Algorithm Comparison

Four algorithms were trained and evaluated on the standardized feature set using $K = 2$ ($K = \text{best_k_silhouette}$), except DBSCAN which determines its own number of clusters from density parameters ($\text{eps} = 0.8$, $\text{min_samples} = 8$).

Algorithm	Silhouette Score	Davies-Bouldin	Clusters
K-Means	0.2985	1.3738	2
Hierarchical (Agglomerative)	0.2445	1.4530	2
Gaussian Mixture Model	0.2767	1.4282	2
DBSCAN	0.2493	1.0825	3 (+ 1,896 noise pts)

Best Algorithm: K-Means — it achieved the highest Silhouette Score (0.2985) among all candidates. DBSCAN produced the lowest (best) Davies-Bouldin Index, but flagged a very large share of the dataset (1,896 of 2,240 records) as noise, making it less suitable as the primary segmentation model for business use.

7.1 Gaussian Mixture Model Details

Metric	Value
Silhouette Score	0.2767
Davies-Bouldin Index	1.4282
AIC	-5201.64
BIC	-4167.37

8. Cluster Profiling (K-Means, Final Model)

Cluster	Size	Avg. Income	Avg. Store Purchases	Avg. Catalog Purchases	Business Label
0	1,235	\$37,699.75	3.64	0.76	Regular Customers
1	1,005	\$70,103.35	8.43	5.00	High Value Customers

Cluster 1 represents higher-income customers with substantially more frequent store and catalog purchases — a natural target for premium offers and loyalty programs. Cluster 0 represents lower-income, lower-frequency shoppers who may respond better to value/discount-driven campaigns.

9. Cluster Stability Analysis

9.1 Random-State Stability (Adjusted Rand Index vs. base model)

Random State	42	100	200	300	500
Adjusted Rand Index	1.00	1.00	1.00	1.00	1.00

A perfect ARI of 1.00 across all tested random states confirms the K-Means solution is fully stable and reproducible.

9.2 Bootstrap Stability (Silhouette Score)

Iteration	1	2	3	4	5
Silhouette Score	0.2931	0.2956	0.2981	0.2989	0.2970

Silhouette scores remain tightly clustered around ~0.295-0.299 across bootstrap resamples, confirming the segmentation is not an artifact of sampling.

10. Final Model Selection

Criterion	Result
Best Algorithm	K-Means
Optimal K	2
Silhouette Score	0.2985
Davies-Bouldin Index	1.3738
Stability (ARI across random states)	1.00 (perfectly stable)

K-Means was selected as the production model based on the combination of the highest Silhouette Score among all four algorithms, strong interpretability (2 clearly separable, business-meaningful segments), and perfect stability under repeated runs and bootstrap resampling.

11. Production Pipeline & Artifacts

The following artifacts were generated and saved for deployment / reuse:

- `models/kmeans_model.pkl`, `gmm_model.pkl` — trained clustering models
- `models/standard_scaler.pkl`, `minmax_scaler.pkl` — fitted scalers
- `pipeline/preprocessing_pipeline.pkl` — feature list + scalers
- `pipeline/customer_segmentation_pipeline.pkl` — full production bundle (features, scaler, best K, all four trained models)
- `reports/` — 20+ CSV reports (validation, correlation, scaling, elbow, silhouette, Davies-Bouldin, evaluations, cluster profiles, stability, comparisons)
- `plots/` — distribution plots, boxplots, pair plot, correlation heatmaps, elbow curve, silhouette curve/plot, DB curve, PCA and t-SNE visualizations, dendrogram, cluster heatmap

12. Business Recommendations

- **High Value Customers (Cluster 1):** Prioritize for premium product lines, early access campaigns, and loyalty rewards given their higher income and purchase frequency across all channels.
- **Regular Customers (Cluster 0):** Target with value-oriented and discount-driven promotions to increase purchase frequency and average order value.
- **Campaign targeting:** Use `Accepted_Campaigns` and channel-specific purchase counts (`web/store/catalog`) to refine channel selection per segment.
- **Model monitoring:** Re-run stability analysis periodically as new customer data accumulates, since segment boundaries may shift over time.

13. Future Improvements

- Test alternative DBSCAN `eps/min_samples` values to reduce the noise ratio and evaluate whether it can produce competitive, usable segments.
- Explore a higher K (e.g. 3-5) with business review, since finer segments may still be actionable despite a lower Silhouette Score.
- Add RFM (Recency-Frequency-Monetary) based scoring as a complementary segmentation lens.
- Automate periodic re-training as new customer data arrives, with drift detection on cluster centers.

Project Status: **Completed Successfully** — All 18 tasks executed.